Lesson 5 – Probability and Statistics Notations

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Basic Probability

Probability of an event A occuring is written as $P(A)$, where:

$$0 \leq P(A) \leq 1$$

Conditional Probability

The conditional probability of event A given B is:

$$P(A \mid B)$$

This is read as *" The probability of A given B."*

Intersection and Union

The probability of both event A and B occuring simultaneously is:

$$P(A \cap B)$$

The probability of either event A or B occuring is:

$$P(A \cup B)$$

Compliment

The compliment of an event A is the probability that A does not occur:

$$p(A^c)$$

This follows the fundamental rule:

$$P(A) + P(A^c) = 1$$

Expectation

Expectation of a random variable X represents its long run average value:

$$\mathbb{E}[x]$$

The conditional expectation of X given Y is:

$$\mathbb{E}[X \mid Y]$$

Variance and Covariance

Variance of a random variable X measures how much it spreads around its mean:

$$\mathrm{Var}(X)$$

Covariance of two random variable X and Y measures how much they vary together.

$$\mathrm{Cov}(X, Y)$$

These two quantities are connected by the identity:

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X)$$

Probability Distributions

A random variable X follows a normal distribution with mean μ and variance σ^2:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Other common distribution is data science:

$$X \sim \mathrm{Bernoulli}(p)$$

$$X \sim \mathrm{Binomial}(n, p)$$

$$X \sim \mathrm{Poisson}(\lambda)$$

Statistical Estimate

An estimator approximates unknown population parameter from sample data. The estimated mean and variance are writen as:

$$\hat{\mu} \quad \text{and} \quad \hat{\sigma}^2$$

In regression analysis, the estimated coefficients and predicted values are writen as:

$$\hat{\beta} \quad \text{and} \quad \hat{y}$$

Bayesian Notation

Baye's theorem describes how to update a believe about parameter θ after observing data $\mathbf{x}$.

$$p(\theta \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \theta)\, p(\theta)}{P(\mathbf{x})}$$

In machine learning literature, this is writen in its compact form as:

$$p(\theta \mid \mathbf{x}) \propto p(\mathbf{x} \mid \theta)\, p(\theta)$$

This is read as posterior is proportional to likelihood times prior.